

MATH20250 HW2 Observations

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General Observations

These are the general, "be careful"-type observations. Some are new from this problem set, while others are repeated.

- **Observations on Proofs**

- (a) **Directionality:** Suppose you have to prove " A if and only if B ." ($A \Leftrightarrow B$) This is a **twofold** process:
 - i. **Prove "if A , then B ."** This may be done by supposing A is true and proving B directly, supposing A is true but B is false and proving by contradiction, or by contrapositive – "if A , then B " is logically equivalent to "if not B , then not A ."
 - ii. **Prove "if B , then A ."** This may be done by supposing B is true and proving A directly, supposing B is true but A is false and proving by contradiction, or by contrapositive – "if B , then A " is logically equivalent to "if not A , then not B ."
- (b) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives.**

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning "is an element of," is generated by `\in`.

- (c) **Existence:** If you are tasked with proving the existence of a quantity, **do not presuppose its existence in your notation.**
 - i. Example: LADW 1.6.7 – The goal is to prove that the matrix B is invertible given that the matrices A and AB are invertible. It may be tempting at some point in the proof to use the following fact from Theorem 6.3 (pg. 26):

$$(AB)^{-1} = B^{-1}A^{-1}.$$

However, there is a key condition attached to that fact:

If linear transformations A and B are **invertible**...

As such, this fact should not be employed unless you have already shown that A **and** B are invertible.

- (d) **Justifications:** There are two main things to remember when justifying the steps in a proof:
- Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly**. If you need to use a result that comes from another homework problem, **prove it explicitly**. Do not take other homework problems for granted.
 - If you are unsure whether to justify something explicitly, **do it**. Better safe than sorry.

- **Miscellaneous**

- Read your matrix dimensions carefully.** Make sure you do not write down a 3×2 matrix when you are working in $M_{2 \times 3}(\mathbb{R})$.
- Check your calculations.** Especially beware of dropping negative signs.
 - If you are finding a solution to a linear system of equations, **substitute your answer to check**. Sometimes, after converting to matrix form, it can be difficult to remember that there was an underlying group of equations. Use them to your advantage.
 - If you are decomposing a matrix (e.g., writing it as a product of elementary matrices), **multiply the resulting matrices** to make sure you get the result you expect.
- Read the problems carefully.** Make sure you answer **every part of the problem**.

Problem-Specific Observations

As with the previous problem set, I have not included are the problems covered in the homework debrief – for those, **please** consult the debrief file on Canvas.

- **LADW 1.5.1 and LADW 2.2.1:** These problems had this boxed "X" next to them: X. This means you should **NOT** choose these problems for your HW submission. In the future, you will not receive credit for problems that have been so marked.
- **LADW 2.3.4 and LADW 2.3.5**

- Spanning set for $\mathbb{P}_3$:** Recall that $\mathbb{P}_3$ is the space of all polynomials of degree *less than or equal to* 3. As such, its basis vectors are 1, x , x^2 , and x^3 ; this gives the following:

$$\mathbb{P}_3 \simeq \mathbb{F}^4 \Rightarrow \dim(\mathbb{P}_3) = \dim(\mathbb{F}^4) = 4.$$

This means that a spanning set for $\mathbb{P}_3$ must contain **at least** four vectors. So, the provided polynomials **do not** span $\mathbb{P}_3$.

- Basis for $\mathbb{F}^4$:** Since $\dim(\mathbb{F}^4) = 4$, a *linearly independent* spanning set for the space must contain **exactly** four vectors; otherwise, the set is linearly dependent. This means there does **not** exist a set of five linearly independent vectors in $\mathbb{F}^4$.